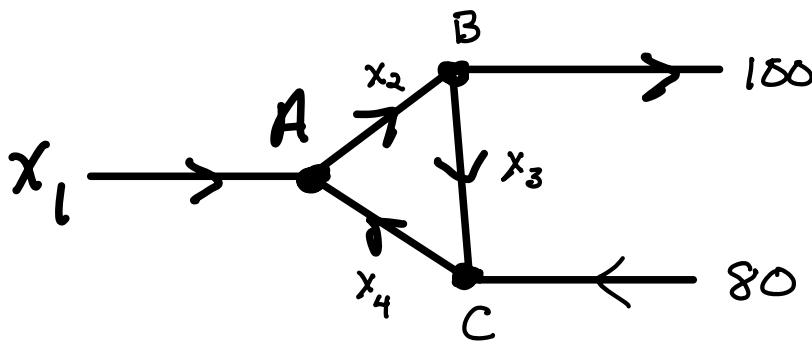


Class 8: Applications

Applications in linear algebra require you to convert a situation into a system of linear equations to be solved and interpreted. The goal of this introductory course is primarily to help you discover the tools that you can use to solve and interpret. You will encounter situations where this type of mathematical modeling will be useful in almost every field of study. Just search on the internet for linear algebra and any other field of study and you will probably get plenty to read.

Example 1: Network Flow

Problem 12 page 55. Find the general flow pattern of the network below. Assuming that the flows are all nonnegative, what is the smallest possible value for x_4



The key idea is that flow into any vertex must equal the flow out of that vertex.

Vertex	Equation	System
A		
B		
C		

Example 2: Mixture Problem

The nutrition facts for a serving of Cheerios and Life cereals is given in the table below. Determine if it is possible to create a mixture of the two cereals to obtain exactly 300 calories, 10 grams of protein, 60 grams of carbs, and 5 grams of fat.

nutrient	Cheerios	Life
Calories	140	160
Protein	5	4
Carbs	29	33
Fat	2.5	2